



TECHNICAL SPECIFICATION DOCUMENT

FOR

FUNDS TRANSFER

## **Introduction**

TeamApt Transfer Service (Multi Entity Account Switch) is an account number-based, online real-time Inter-Bank Funds Transfer product.

## **Transaction Types**

1. Name Enquiry.
2. Fund Transfer.
3. Transaction Status Query.

## **Interface specification**

TeamApt transfer service is a rest service with json, messages are encrypted using PGP protocol.

The Sandbox URLs as follows:

### Base URLs

- <https://aptpay-account-transfer.switch-staging.teamapt.com/eft/v1> [Internet](#)
- TBD - VPN

\*Production base URL will be provided upon completion of integration and will be only accessible on VPN.

Please note that portal access is on this url.

- <https://aptpay-account-portal.switch-staging.teamapt.com/login>

### Endpoint URLs

- a. [/ne](#) -Name Enquiry
- b. [/ft](#) - Funds Transfer
- c. [/tsq](#) - Transaction status query
- d. [/inst/balance](#) - Institution Balance.

## **Sample Messages**

The sample message for Name Enquiry, Funds Transfer, Status Query & Institution Balance are shown below.

### **Name Enquiry Request**

```
{
  "uniqueReference": "APT00000250117073316818161131413",
  "destinationInstitutionCode": "000000",
  "beneficiaryAccountNumber": "5252525252"
}
```

### **Name Enquiry Response:**

```
{
  "uniqueReference": "APT00000250117073316818161131413",
  "destinationInstitutionCode": "000000",
  "beneficiaryKycLevel": "1",
  "beneficiaryBankVerificationNumber": "77777777777",
  "responseCode": "00",
  "beneficiaryAccountNumber": "5252525252",
  "beneficiaryAccountName": "Maximus Decimus Meridius"
}
```

Fund Transfer Request

```
{  
  "uniqueReference": "APT00000250117073422231121121322",  
  "destinationInstitutionCode": "000000",  
  "amount": "1,000",  
  "nameEnquiryReference": "APT00000250117073316818161131413",  
  "beneficiaryKycLevel": "1",  
  "beneficiaryBankVerificationNumber": "1111111111",  
  "originatorBankVerificationNumber": "0909090909",  
  "paymentReference": "xua7d19eL4",  
  "narration": "Transfer from Emulator to Test",  
  "beneficiaryAccountNumber": "5252525252",  
  "beneficiaryAccountName": "Maximus Decimus Meridius",  
  "originatorAccountName": "Simulator",  
  "originatorAccountNumber": "1111111111",  
  "originatorKycLevel": "1"  
}
```

Funds Transfer Response:

```
{
  "uniqueReference": "APT00000250117073422231121121322",
  "destinationInstitutionCode": "000000",
  "amount": "1000",
  "nameEnquiryReference": "APT00000250117073316818161131413",
  "beneficiaryKycLevel": "1",
  "beneficiaryBankVerificationNumber": "1111111111",
  "originatorBankVerificationNumber": "0909090909",
  "paymentReference": "xua7d19eL4",
  "responseCode": "00",
  "narration": "Transfer from Emulator to Test",
  "beneficiaryAccountNumber": "5252525252",
  "beneficiaryAccountName": "Maximus Decimus Meridius",
  "originatorAccountName": "Simulator",
  "originatorAccountNumber": "1111111111",
  "originatorKycLevel": "1"
}
```

Transaction Status Query Request

```
{  
  "uniqueReference": "APT00000250117073422231121121322"  
}
```

Transaction Status Query Response

```
{  
  "uniqueReference": "APT00000250117073422231121121322",  
  "responseCode": "00"  
}
```

Institution Balance Request – (For institutions using balance model)

JSON

```
{  
  "type": ""  
}
```

Institution Balance Response

JSON

```
{
  "amount": "1000000000.00",
  "responseCode": "00"
}
```

**Note:** These are expected to be encrypted first, then converted to hexadecimal before transmission to TeamApt.

Sample Encryption Message/Data.

Request:

```
{
  "request": "85010C031C103C3BBE79B7570107FD15FC8909BAEEFE9FFB69CF2441B4B24AFDBEDE7AC6222224FB296F
876875841D47FA242B29ED17B1EE4D279F9149D5B8EB33610ACE540880DF2BAC2D4838CECEF54274DB149C1C880C9
7402561DF5EA8F72CD56C161C18A55AD62AAE4D98AE0CB17A88A24353203DC912990A206A1B01BA98B59AC97E979
CE1195BDAFFD1F0B2EE34DEEE6E2076E3A759720780C463F9972DE7C075FB4CB8B070935AF5577DCC75BC3E0E1C6D
A7E7FDCE376EEFD7CCBBA1517C8DACCA6DE0D8EDEDCC8A54D4F7D864E6460704D07E1142498FF9A553B2995150D6
6E3600D3F3CB4963C7DADA01380BE1CF95F6FA84A9CAAA4B58149C7FD13FC24CA0B4C8B5275FD372090647FD2C07
0018082EC25776F71003F03E9CD83E0AC5C2DD927B345C2C0F59C0A4DBFD025D1F847B1E2A1EFA73EE4DF3984F86
49B8C8498771AD65A411C8AB397219044BA0B62A50B14367EC8115B80E9DDF1E739097F195297C72B590E35C91CC3
222EE5820E2D361E469024526CA8A13273AF1FD1840031085390D9DDF5F23D2D2DBD8D53DEA27FDD788DDF2B541
08C5D248EAA22CBD883DA2D411DBD188495CCBE0D11B72B73D8ADCDCFEA9D973F77A5E4FA448C7529563884F734
F02F638BA0CDBDE1473B25BA439936C11774B2BBC43597410F5576193B5D4DE528620D36A165C9B5BD914C3087CD
EBAB301B09E61D0CCBB3C98757B58212991CFA8952BB8FD5A94A5F41AC893693D973F5458C52DC5AEF055EB7476EE
5EB88511F724D63191826D5CFD278FDA7FB6DDFF079118817491961;"
}
```

the "request" field is the encrypted JSON request data

Response:

```
{
  "statusCode": "90000",
  "message": null,
  "data": "85010C031C103C3BBE79B7570108008353E3D1D05341F17D59D1184AA3BC723D5C57FED9C010C48FEDACE6
CD0577A7DF6AE0865648C20C6AE732E3C5F02A50029880B12ECDECFD3D8E29D8502C010541E8BB76CEA7EF3CA432F
AE767562B2B9D8F4CEFE5B65B68DE74F36029DA038D255BD739B410FE29B086773227A73387BE9A4A8B30613586D6
5E78BDFA7F0CEAF5752C6006DA7924C613D8636335E7027EC67B0607147101977B8EBD42DEB4F6F3DBBA96CBA9B4C
DEF8626A62CE016A2431A60850AB1F8934A29AA2690FAA164800DA8379B631AD0905E6666AB268720BA7FEB57706
0316735ED83629EE2B68D7E5BA958E7020D8FD0E550D24CFEB55765CA663D8E3C5718DC76529E7A23AF72D2C0F401
090F8A3E750C451823BCB935112800B7D11770EDE566F79316E6358BC53BC3DAE3D0C7C15CB006BB85B858AB9469D
B077B35BB5F4DAC36B12AE204D560B89DBC377A980133CD3DF706DB19B6B2EE27828C31C6DA62F9ACE54B6DAF90B
85C26E94E2C8D08D9E13C5872BF0C49CFB1C42320FC565AE0C26A70EBED6F61C21BC2C4C4B602640190F135C67641
7093E7E0622137B6EDC5A33F3B667C777C04BC4217F81C6D581D8D3D3DF6A5BAC839F128C2418541713C1A056AE3
```

```

D0C823E0C3049947B48258BE56F4157AC804B65671F12BC69F726BF29AA5C6904ADE82C4DFD5327E8242237C50FDC
6DD60F558FE08C114907846F8FEB33470D16CB3A10528992079D2D9863213F73744B144E776B693D1614528BD50B2
1F1A250C9825F906B2EE09E0E48E260769BCFEA3F59A78DB12A9772B2C7EE16F0FF277C2D6349CDDA2BCC9E9444D5
EE047D95A4F8356FE06AE13080DDD20B6A8BDA476EAD1AB6148EB6D7E62FFBCE9C20B6BDF3F944CCA110F7082D56
5D78D81C04A2D0410092CEE327E948610B0BD43B9C9DED0AB15B683A0F3F0A147DB11EA900D6E4384FEF2DECC0B8
639A708E26443703122E4B6F5279AE4353D1B;"
}

```

the "data" field is the encrypted JSON response data

## **Field Descriptions**

| <b>S/N</b> | <b>Field</b>               | <b>Description</b>                                                                                                                                                                                                                                                                       | <b>Length</b> |           |
|------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|
| 1.         | uniqueReference            | <p>Unique number that identifies transactions in the following format:</p> <p>Char 1 – 8: Sender's TeamApt's Prefix</p> <p>Char 9 – 20: Date and time (format yyMMddHHmmss - HH is 24-hour clock)</p> <p>Char 21 – 32: 12 – character unique number (either serial or random number)</p> | 32            | Mandatory |
| 2.         | destinationInstitutionCode | Destination Institution's NIBSS Code                                                                                                                                                                                                                                                     | 6             | Mandatory |
| 3.         | responseCode               | Code showing the status of transaction                                                                                                                                                                                                                                                   | 2             | Mandatory |
| 4.         | nameEnquiryReference       | Unique reference of the name enquiry done prior to a fund transfer                                                                                                                                                                                                                       | 32            | Mandatory |



|     |                                   |                                                                          |                  |           |
|-----|-----------------------------------|--------------------------------------------------------------------------|------------------|-----------|
| 5.  | beneficiaryAccountName            | Beneficiary Account Name of financial institution's customer             | Maximum of 100   | Mandatory |
| 6.  | beneficiaryAccountNumber          | Beneficiary Account Number of financial institution's customer           | Maximum of 20    | Mandatory |
| 7.  | beneficiaryBankVerificationNumber | Beneficiary Bank Verification Number                                     | Maximum of 20    | Mandatory |
| 8.  | originatorAccountName             | Originator Account Name of financial institution's customer              | Maximum of 100   | Mandatory |
| 9.  | originatorAccountNumber           | Originator Account Number of financial institution's customer            | Maximum of 20    | Mandatory |
| 10. | originatorBankVerificationNumber  | Originator Bank Verification Number                                      | Maximum of 20    | Mandatory |
| 11. | originatorKycLevel                | Originator's Know Your Customer Level                                    | 1                | Mandatory |
| 12. | narration                         | Description of transaction                                               | Maximum of 100   | Mandatory |
| 13. | paymentReference                  | Reference of Payment                                                     | Maximum of 100   | Mandatory |
| 14. | amount                            | Value of fund in naira                                                   | 2 decimal places | Mandatory |
| 15. | beneficiaryKycLevel               | Know Your Customer Level of customer's account that is to receive credit | 1                | Mandatory |

## **Response Codes**

| Code | Description | Comment |
|------|-------------|---------|
|------|-------------|---------|

|    |                                     |                                                                    |
|----|-------------------------------------|--------------------------------------------------------------------|
| 00 | Approved or completed successfully  | Final Status – Request was successful                              |
| 05 | Do not honor                        | Final Status – Request FAILED                                      |
| 06 | Dormant Account                     | Final Status – Request FAILED                                      |
| 07 | Invalid Account                     | Final Status – Request FAILED                                      |
| 08 | Account Name Mismatch               | Final Status – Request FAILED                                      |
| 09 | Request Pending                     | **Status <b>NOT</b> Final – Re-query till final status is received |
| 12 | Invalid transaction                 | Final Status – Request FAILED                                      |
| 13 | Invalid Amount                      | Final Status – Request FAILED                                      |
| 15 | Invalid Unique Reference            | Final Status – Request FAILED                                      |
| 16 | Unknown Beneficiary Bank Code       | Final Status – Request FAILED                                      |
| 25 | Unable to locate record             | Final Status – Request FAILED                                      |
| 26 | Duplicate record                    | Final Status – Request FAILED                                      |
| 30 | Format error                        | Final Status – Request FAILED                                      |
| 57 | Transaction not permitted to sender | Final Status – Request FAILED                                      |
| 61 | Transfer limit Exceeded             | Final Status – Request FAILED                                      |
| 63 | Security violation                  | Final Status – Request FAILED                                      |
| 91 | Beneficiary Bank not available      | Final Status – Request FAILED                                      |
| 92 | Routing error                       | Final Status – Request FAILED                                      |
| 96 | System malfunction                  | Final Status – Request FAILED                                      |
| 97 | Timeout                             | **Status <b>NOT</b> Final – Re-query till final status is received |

**\*SLA for final status is 6 minutes**

## **Message Security**

The encryption is based on PGP RSA technology. All messages will be encrypted with TeamApt's public key, which can be downloaded via the web portal. Subsequently, response messages from TeamApt will be encrypted with the sending institution's public key. The sending institution's public key should be uploaded via the web portal.

All requests should include the sender's API key in the http header as follows:

*x-api-key: <apiKey>*

The API key can be retrieved from the web portal.

## **Reports**

TeamApt shall make available transaction reports to both the sender's and beneficiary's financial institution at the end of every settlement cycle. The reports would be made available on the operational web-based TeamApt Transfer Service Platform.

## **Dispute Resolution**

Dispute Resolution will be based on the Procedure below:

- a. The Sender lodges complaint(s) with his/her financial institution.
- b. The Sender's financial institution performs first level check such as:
  - Authenticate Sender
  - Confirms that account was debited
  - Checks for auto-reversal etc
- c. Based on the first – level check ascertaining that the dispute is indeed genuine, the sender's financial institution logs the issue on the TeamApt Transfer Service Dispute Management Capability.
- d. TeamApt Transfer Service Dispute Management Capability stores the disputes.
- e. The Beneficiary's financial institution receives and verifies disputes from the transaction traces and logs.
- f. The Beneficiary's financial institution through the TeamApt Transfer Service Dispute Management Capability responds with a 'Accepted' or 'Declined'.
- g. If response is not received from the Beneficiary's financial institution within 72 hours, TeamApt would recover the contested fund automatically from the Beneficiary's financial institution back to the Sender's financial institution.
- h. If the response from the Beneficiary's financial institution is 'complaint invalid', no further action will be taken. However, if there is a repeat lodgment by the Sender's financial institution of this complaint, it would be referred to Arbitration Panel

(comprising representatives of the two banks involved and TeamApt) for resolution and the decision of the Panel will be final.

- i. If the response from the Beneficiary's financial institution is 'Accepted', it would notify the Dispute capability which, based on this instruction, recover funds from the Beneficiary's financial institution to the Sender's institution for final disbursement to the Sender.